

Inference for instrumental variables

Maximilian Kasy

Department of Economics, University of Oxford

Fall 2025

Outline

- Asymptotics of reduced form parameters.
- Delta method for the Wald estimator.
- Weak instruments and failure of the delta method.
- Anderson-Rubin test and confidence sets

Takeaways for this part of class

- Standard asymptotics give an approximate normal distribution for the Wald estimator.
- This is a bad approximation in the case of weak instruments: The remainder of a first-order expansion is too large.
- Valid inference is still possible, using likelihood ratio tests.

Wald estimator

- Recall the 2SLS / Wald estimand:

$$\beta = \frac{\alpha}{\gamma},$$

where

$$\alpha = \text{Cov}(Z, Y), \quad \gamma = \text{Cov}(Z, D).$$

- Sample analog:

$$\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}},$$

where

$$\hat{\alpha} = \text{Cov}_n(Z, Y), \quad \hat{\gamma} = \text{Cov}_n(Z, D).$$

Asymptotic distribution of the reduced form

For some variance matrix Σ ,

$$\sqrt{n} \cdot \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \rightarrow N(0, \Sigma)$$

Practice problem

- Prove this, using the law of large numbers and the central limit theorem.
- You can assume $E[Z] = 0$ and $\hat{\alpha} = E_n[Z \cdot Y]$, $\hat{\gamma} = E_n[Z \cdot D]$ for simplicity.

Sketch of solution

- Let $W_i = (Z_i Y_i, Z_i D_i)$.
- The multivariate central limit gives

$$\frac{1}{\sqrt{n}} \cdot \sum_i (W_i - E[W_i]) \rightarrow N(0, \Sigma),$$

where $\Sigma = \text{Var}(W_i)$.

- The claim follows.
- For the general case ($E[Z] \neq 0$), the delta-method yields the same result.

Asymptotic distribution of the Wald estimator

Let again $\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}}$, where $\hat{\alpha} = \text{Cov}_n(Z, Y)$, $\hat{\gamma} = \text{Cov}_n(Z, D)$.

Then, for some σ^2 ,

$$\sqrt{n} \cdot (\hat{\beta} - \beta) \rightarrow N(0, \sigma^2).$$

Practice problem

- Prove this, using
 - asymptotic normality of the reduced form,
 - and the delta method.
- What is σ^2 ?

Sketch of solution

- Let $f(a, c) = \frac{a}{c}$, so that $\hat{\beta} = f(\hat{\alpha}, \hat{\gamma})$.
- Consider the first-order Taylor approximation $f(a, c) \approx f(\alpha, \gamma) + df \cdot \begin{pmatrix} a - \alpha \\ c - \gamma \end{pmatrix}$,
where $df = \left(\frac{\partial}{\partial a} f(\alpha, \gamma), \frac{\partial}{\partial c} f(\alpha, \gamma) \right) = \left(\frac{1}{\gamma}, -\frac{\alpha}{\gamma^2} \right)$.
- Recall that $\sqrt{n} \cdot \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \rightarrow N(0, \Sigma)$.
- The delta method gives $\sqrt{n} \cdot (f(\hat{\alpha}, \hat{\gamma}) - f(\alpha, \gamma)) \rightarrow N(0, \sigma^2)$, where $\sigma^2 = df \cdot \Sigma \cdot df'$.

Not so fast!

Suppose the asymptotic distribution of the reduced form is exact, so that

$$\begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \sim N\left(0, \frac{1}{n}\Sigma\right)$$

Practice problem

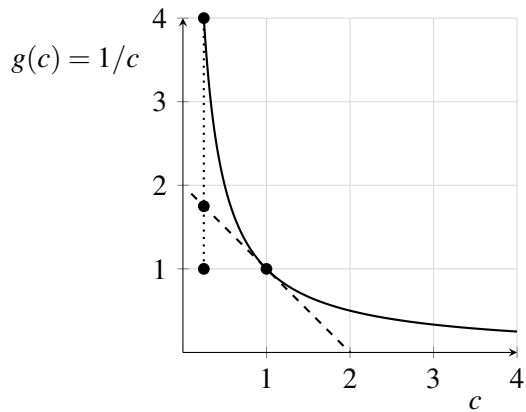
- Under this assumption, is $\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}}$ unbiased for β ?
- Does the expectation even exist?
- Hint: Consider the special case of $\alpha = \beta = 0$, and look up the Cauchy distribution.

When the delta-method fails

- The key step in our asymptotic normal approximation for $\hat{\beta}$ was a first-order approximation to the function $f(a, c) = \frac{a}{c}$,
$$f(a, c) \approx f(\alpha, \gamma) + \frac{\partial}{\partial a} f(\alpha, \gamma) \cdot (a - \alpha) + \frac{\partial}{\partial c} f(\alpha, \gamma) \cdot (c - \gamma).$$
- First-order approximations are poor when the function is very non-linear.
- For intuition, consider a first-order approximation to the function $g(c) = \frac{1}{c}$,
$$g(c) \approx \tilde{g}(c) = \frac{1}{\gamma} - \frac{1}{\gamma^2} \cdot (c - \gamma).$$
- This approximation gets infinitely bad when c is close to 0:

$$\frac{g(c) - g(\gamma)}{\tilde{g}(c) - \tilde{g}(\gamma)} = \frac{\frac{1}{c} - \frac{1}{\gamma}}{-\frac{1}{\gamma^2} \cdot (c - \gamma)} = \frac{\frac{1}{\gamma c} \cdot (c - \gamma)}{\frac{1}{\gamma^2} \cdot (c - \gamma)} = \frac{\gamma}{c}.$$

Illustration



Testing null hypotheses on β

- Consider the null hypothesis $\beta = \frac{\alpha}{\gamma} = \beta_0$.
- Equivalently, $\alpha - \gamma \cdot \beta_0 = 0$.
- Under this null hypothesis,

$$\hat{\alpha} - \hat{\gamma} \cdot \beta_0 \sim N(0, \tau_{\beta_0}^2),$$

where

$$\tau_{\beta_0}^2 = \Sigma_{\alpha, \alpha} - 2\beta_0 \cdot \Sigma_{\alpha, \gamma} + \beta_0^2 \cdot \Sigma_{\gamma, \gamma}.$$

- This suggests a t-test which rejects if

$$\left| \frac{\hat{\alpha} - \hat{\gamma} \cdot \beta_0}{\hat{\tau}_{\beta_0}^2} \right| > z_{.975}.$$

- This is known as the *Anderson-Rubin* test.

Confidence sets using test inversion

- Like any test, the Anderson-Rubin test can be inverted to give a confidence set for β :

$$CI = \left\{ \beta_0 : \left| \frac{\hat{\alpha} - \hat{\gamma} \cdot \beta_0}{\hat{\tau}_{\beta_0}^2} \right| \leq z_{.975} \right\}$$

- Note that CI is not necessarily a connected interval:
 - For $\frac{\hat{\gamma}}{\sqrt{\Sigma_{\gamma,\gamma}}}$ close to 0, $CI = (-\infty, \infty)$.
 - For $\frac{\hat{\gamma}}{\sqrt{\Sigma_{\gamma,\gamma}}}$ large, CI is approximately the delta-method based interval.
 - In intermediate cases, $CI = (-\infty, a] \cup [b, \infty)$.

References

- *Andrews, I., Stock, J., and Sun, L. (2024). Weak instruments in IV regression: Theory and practice. Annual Review of Economics.*
- *Kasy, M. (2019). Uniformity and the delta method. Journal of Econometric Methods, 8(1).*